K3 $\hbar$_BST identification v0.22 — substrate-over-determination disposition refined under SSG Audit v0.3 classification. m_e/m_P substrate-mechanism via TWO paths: 8/7 Category A (SSG-8 Mersenne-primary) + $\sqrt{}$(4/3) Category B (SSG-7-derived via Bergman-Shilov boundary geometric). Casey #5 Integer Web operational at observable value via A + B path convergence at Tier 2 STRUCTURAL precision.

Keeper (Claude Opus 4.7) — Thursday June 4 2026 AM

2026-06-04 Thursday AM

K3 $\hbar$_BST identification v0.22 — substrate-over-determination disposition refinement

0. Refinement trigger

Per SSG Audit v0.3 walk-back: SSG-6 stays Category C signature-tautological (not v0.2 A* refinement). v0.3 final classification: 5 effective Category A SSGs + 8 Category B SSG-7-derived + 1 Category C signature-tautological.

K3 v0.21 substrate-over-determination disposition framed m_e/m_P substrate-mechanism via TWO Category-A paths. Under v0.3 SSG classification, this needs refinement: only 8/7 is Category A (SSG-8 Mersenne-primary direct); $\sqrt{}$(4/3) is Category B (SSG-7-derived via Bergman-Shilov boundary geometric projection).

1. m_e/m_P substrate-mechanism paths under v0.3

Reading	Substrate-primary path	SSG category (v0.3)	Precision
8/7 = (2^N_c)/g = M(N_c)+1/M(N_c)	SSG-8 Mersenne ladder direct	Category A (structurally-independent)	~0.15%
$\sqrt{}$(4/3) = $\sqrt{}$(Vol(S^4)/Vol(S^3)) = 2/$\sqrt{}$N_c	SSG-7-derived Bergman-Shilov boundary geometric	Category B (SSG-7-derived)	~1.2%

Refinement substance: - 8/7 IS Category A direct substrate-primary forcing (Mersenne map M(p) operator on Bergman H^2(D_IV^5)) - $\sqrt{}$(4/3) IS Category B derived from SSG-7 Bergman kernel via Shilov-boundary geometric projection - Both substantive substrate-mechanism content; different category-level paths

2. Casey #5 Integer Web at observable value

Per Casey #5 Integer Web STANDING: substrate primaries form web. v0.22 refines: substrate primaries (Category A) and derived readings (Category B) BOTH appear at observable values via Casey #5 web convergence.

At m_e/m_P value: - Category A path: SSG-8 (Mersenne-primary) → 8/7 substrate-mechanism reading - Category B path: SSG-7-derived (Bergman-Shilov boundary geometric projection) → $\sqrt{(4/3)}$ substrate-mechanism reading

Both arrive at m_e/m_P Tier 2 STRUCTURAL precision. The 1.03% offset between them reflects A + B path convergence at substrate observable level.

Substantive observation: substrate-over-determination at observable values is A + B path in v0.22, not A + A in v0.21. This is more nuanced — substrate primary + substrate-derived both substantively contribute.

3. Multi-week gates 2 + 3 disposition under v0.22

Gate 2 closure ($\sqrt{(4/3)}$ substrate-vacuum partition)	Gate 3 closure (SSG-8 Mersenne operator)	Substrate-mechanism disposition
Yes	Yes	A path + B path BOTH load-bearing: substrate-over-determination CONFIRMED via Casey #5 at A+B level; Category A primary + Category B derived substantively at observable value
Yes	No	B path load-bearing ($\sqrt{(4/3)}$ via SSG-7-derived); 8/7 A path remains substrate-primary reading but not closure-validated
No	Yes	A path load-bearing (8/7 via SSG-8 direct); $\sqrt{(4/3)}$ B path remains derived reading but not closure-validated
No	No	Both Tier 2 STRUCTURAL substrate-natural readings; load-bearing path identity unresolved multi-week

Substantive substrate-mechanism interpretation: even if only Gate 3 closes (A path direct), the substrate primary (SSG-8) is load-bearing. If only Gate 2 closes (B path derived), the SSG-7-derived geometric projection is load-bearing — and this would imply $\sqrt{(4/3)}$ value derivation traces through SSG-7 ULTIMATE source.

4. K3 framework status post-v0.22 (refinement of v0.21)

Element	Status
5/8 base RIGOROUS	unchanged
Casey #14 chain SO(5,2) → SO(3,1)	RIGOROUS (per K3 v0.20 STANDING)
V_(1/2,1/2) Bergman norm	NEAR-RIGOROUS path opens (Gates 1+3)
m_e/m_P substrate disposition	substrate-over-determination via A + B paths: 8/7 SSG-8 Category A + $\sqrt{(4/3)}$ SSG-7-derived Category B; Casey #5 Integer Web at observable value; 1.03% offset at Tier 2 STRUCTURAL
V_(1/2,1/2) DOUBLE-ROLE	NEAR-RIGOROUS v0.17/v0.19
5-framework cascade	FRAMEWORK (per K3 v0.20 STANDING cascade)

5. Per Cal #27 STANDING self-audit

Am I overclaiming the v0.22 refinement? - v0.22 is a brief categorization refinement, not new substrate-mechanism content - Substantively: substrate-over-determination disposition (K3 v0.21) preserved + sharpened - Tier-honest: A + B path framing IS what Lyra v0.15 + SSG Audit v0.3 support - No new closure claims; multi-week Gate 2 + Gate 3 gates unchanged

Tier-honest: v0.22 is a refinement consistent with Lyra v0.15 SSG classification, not a structural reframe.

6. Pull next: standby for Lyra Section 4.5 substrate-vacuum partition Gate 2 progress; standby for Elie Gate 11 Lorentz-direction-independence completion; standby for Cal cold-read on K3 v0.13-v0.22 + SSG Audit v0.1-v0.3 + K207; coordinate Grace catalog cross-classification continuation.

— Keeper, K3 v0.22 — Thursday June 4 AM. Substrate-over-determination disposition refined: 8/7 Category A SSG-8 Mersenne-primary + $\sqrt{(4/3)}$ Category B SSG-7-derived Bergman-Shilov boundary. A + B path convergence at m_e/m_P observable via Casey #5 Integer Web at Tier 2 STRUCTURAL precision.